



Hamiltonian description of toroidal magnetic fields in vacuum

Jason W. Bates and H. Ralph Lewis

Citation: [Physics of Plasmas \(1994-present\)](#) **4**, 2619 (1997); doi: 10.1063/1.872551

View online: <http://dx.doi.org/10.1063/1.872551>

View Table of Contents: <http://scitation.aip.org/content/aip/journal/pop/4/7?ver=pdfcov>

Published by the [AIP Publishing](#)

Articles you may be interested in

[Analytical description of stochastic field-line wandering in magnetic turbulence](#)

Phys. Plasmas **14**, 092903 (2007); 10.1063/1.2776905

[Asymptotic, closed integral solutions for the reconstruction of Grad–Shafranov equilibria in axisymmetric, large-aspect-ratio toroidal plasmas](#)

Phys. Plasmas **11**, 186 (2004); 10.1063/1.1625372

[Hamiltonian guiding center equations in toroidal magnetic configurations](#)

Phys. Plasmas **10**, 573 (2003); 10.1063/1.1544500

[Determination of the vacuum field resulting from the perturbation of a toroidally symmetric plasma](#)

Phys. Plasmas **6**, 2781 (1999); 10.1063/1.873235

[Equations for the evolution of the radial electric field and poloidal rotation in toroidally symmetric geometry](#)

Phys. Plasmas **5**, 763 (1998); 10.1063/1.872762



AIP | Journal of
Applied Physics

Journal of Applied Physics is pleased to
announce **André Anders** as its new Editor-in-Chief

Hamiltonian description of toroidal magnetic fields in vacuum

Jason W. Bates and H. Ralph Lewis

Department of Physics and Astronomy, Dartmouth College, Hanover, New Hampshire 03755-3528

(Received 4 February 1997; accepted 31 March 1997)

A Hamiltonian description of vacuum magnetic fields in toroidal geometry is presented. Using a magnetic scalar potential in conjunction with Boozer's canonical representation of a magnetic field [A. Boozer, *Phys. Fluids* **26**, 1288 (1983)], a field-line Hamiltonian for general nonaxisymmetric vacuum fields has been derived. © 1997 American Institute of Physics. [S1070-664X(97)01107-5]

I. INTRODUCTION

The topology of magnetic field lines plays an important role in magnetic fusion research, and magnetic field lines provide examples of Hamiltonian dynamical systems that display integrable and chaotic behavior with rich topological structures. Because magnetic fusion devices are generally toroidal, there is particular interest in the topological properties of the field lines of toroidal magnetic fields. The existence or nonexistence of nested toroidal magnetic surfaces in the plasma confinement system influences the behavior of the plasma significantly.

In the present paper we are concerned with techniques being developed for studying the topology of toroidal magnetic fields. The techniques being developed are based on Boozer's representation of the magnetic field,¹ in which the differential equations that govern the trajectories of magnetic field lines are derivable from Hamiltonian functions $H(\theta, \psi, \phi)$, where θ is a poloidal angle variable, ψ is a toroidal flux variable, and ϕ is a toroidal angle variable (the independent variable). The Hamiltonian H is a poloidal flux function. This representation is applicable when the toroidal component of the field does not vanish. Boozer's representation of toroidal magnetic fields is similar to some earlier formulations; see, for example, Kruskal and Kulsrud,² Greene and Johnson,³ and Hamada.⁴

If the magnetic field is axisymmetric, an autonomous Hamiltonian can be used (i.e., the Hamiltonian does not depend on the variable ϕ). In that case, the Hamiltonian $H(\theta, \psi)$ is a field-line invariant, which means that magnetic field lines lie on the surfaces $H = \text{const}$. These surfaces can be determined by integrating the field-line differential equations to follow the field lines for many revolutions around the toroidal axis. However, they can also be determined by computing the level curves of $H(\theta, \psi)$, without integrating the differential equations. When the magnetic field is not axisymmetric, but a field-line invariant $I(\theta, \psi, \phi)$ is known that is periodic in ϕ , a similar procedure can be used to study the field-line topology. Magnetic surfaces exist at a particular value of ϕ if the invariant is a periodic function of its third argument: $I(\theta, \psi, \phi + 2\pi) = I(\theta, \psi, \phi)$. In that case, the intersection of a magnetic surface with the surface $\phi = \text{const}$ is the curve in the (θ, ψ) -plane given by $I(\theta, \psi, \phi) = \text{const}$.

The idea of seeking field-line invariants for nonaxisymmetric magnetic fields and using them to study field-line topology was discussed by Lewis⁵ in connection with a representation of nonautonomous Hamiltonians in terms of an

invariant.⁶ Lewis specialized the formalism to the case of vacuum magnetic fields;⁵ Abraham-Shrauner and Lewis considered some aspects of the formalism for force-free magnetic fields.⁷

In order to use field-line invariants for studying field-line topology in a plasma, it is first necessary to find a set of variables (θ, ψ, ϕ) and a Hamiltonian $H(\theta, \psi, \phi)$ that describe the magnetic field in some spatial region in the context of a definite plasma model. In one class of models, the region containing plasma is surrounded by a vacuum region. We have found a set of variables (θ, ψ, ϕ) and a Hamiltonian $H(\theta, \psi, \phi)$ in terms of which the field can be represented in a toroidal vacuum region, outside of which there may be an arbitrary current distribution. The variables (θ, ψ, ϕ) are determined self-consistently along with the Hamiltonian. In carrying out the analysis, we use toroidal coordinates (ξ, η, φ) , which are closely related to variables defined by Morse and Feshbach.⁸ (See Appendix A.) Specification of the variables (θ, ψ, ϕ) and the Hamiltonian $H(\theta, \psi, \phi)$ is the subject of this paper.

Although a vacuum field-line Hamiltonian does not provide a self-consistent description of a plasma/magnetic field system, it does allow an investigation of a class of interesting toroidal magnetic fields. The Hamiltonian for vacuum magnetic fields can also serve as the zeroth-order starting point for a perturbative determination of a Hamiltonian for magnetic fields in a more realistic plasma model [e.g., force-free fields or fields in a magnetohydrodynamic (MHD) equilibrium].

We also wish to mention that the Hamiltonian description of magnetic field lines presented here has served as the basis of a recent topological analysis of selected magnetic field configurations using field-line invariants.⁹ In that study, the field-line invariants were approximated by applying a novel perturbation theory recently developed by Lewis, Bates and Finn¹⁰ for constructing invariants of nonautonomous Hamiltonian systems.

This paper is organized as follows. In Sec. II, we review the general Hamiltonian description of nonaxisymmetric, toroidal magnetic fields as discussed by Boozer¹ and Lewis.⁵ Limiting our attention to vacuum magnetic fields in Sec. III, we present a magnetic scalar potential for a toroidal region surrounded by current. Using this scalar potential, we then derive an expression for a nonaxisymmetric, vacuum field-line Hamiltonian. As an example of the Hamiltonian formu-

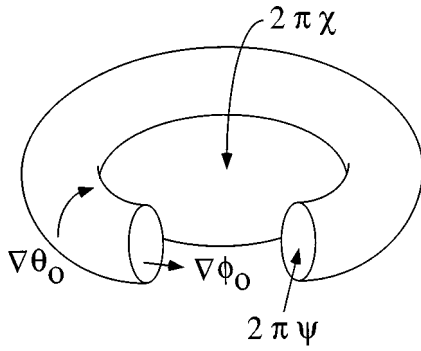


FIG. 1. Canonical variables in toroidal geometry (after Boozer¹).

lation of magnetic field lines, Sec. IV contains an analysis of a simple system: an axisymmetric current loop. Conclusions and a few remarks about subsequent research are given in Sec. V. General differential relations for describing magnetic fields in terms of canonical variables (θ, ψ, ϕ) appear in Appendix B. Appendix C contains derivations of the vector and scalar potentials for a current loop in toroidal coordinates. Throughout our discussion, all field quantities are expressed in terms of Gaussian cgs units.

II. HAMILTONIAN FORMULATION

The Hamiltonian character of the differential equations that describe magnetic field lines has been discussed by many authors.^{11–15} Our starting point is the pioneering work of Boozer,¹ which states that any magnetic field with toroidal topology can be represented in the canonical form

$$\mathbf{B} = \nabla \psi_o \times \nabla \theta_o + \nabla \phi_o \times \nabla \chi_o, \quad (1)$$

where $\theta_o(\mathbf{r})$ is a poloidal angle function, $\phi_o(\mathbf{r})$ is a toroidal angle function, $\chi_o(\mathbf{r})$ is a poloidal flux function, and $\psi_o(\mathbf{r})$ is a toroidal flux function. (Here, the subscript “o” is used to remind us that these quantities are functions of position; when the subscript is absent, we are referring to the value of the function.) If $\mathbf{r}$ returns to its original value by moving along a path that encircles the poloidal axis once, then θ changes by 2π ; if $\mathbf{r}$ returns to its original value by moving along a path that encircles the toroidal axis once, then ϕ changes by 2π . Yoshida¹⁶ has proved on a rigorous mathematical basis that every solenoidal field in an arbitrary smooth simple toroidal domain can be written in this form. The quantities ψ, θ, ϕ and χ are illustrated in Fig. 1. Note that any $\mathbf{B}$ expressed in the canonical form is manifestly divergence-free, $\nabla \cdot \mathbf{B} = 0$, since we can rewrite Eq. (1) in the form $\mathbf{B} = \nabla \times (\psi_o \nabla \theta_o - \chi_o \nabla \phi_o)$, and the divergence of a curl is always zero.

In order to interpret the values of ψ and χ , we define a toroidal flux Φ_{tor} and a poloidal flux Φ_{pol} :

$$\Phi_{tor} = \int_{S_{tor}} \mathbf{B} \cdot d\mathbf{a}, \quad \Phi_{pol} = \int_{S_{pol}} \mathbf{B} \cdot d\mathbf{a},$$

where S_{tor} is a surface whose normal is antiparallel to $\nabla \phi_o$ and which is bounded by a surface $\psi_o(\mathbf{r}) = \psi = \text{const}$; S_{pol} is a surface whose normal is parallel to $\nabla \theta_o$ and which is

bounded by a surface $\chi_o(\mathbf{r}) = \chi = \text{const}$. Then, making use of Eq. (1) and applying Stoke’s theorem we find

$$\Phi_{tor} = \int_{S_{tor}} \nabla \times (\psi_o \nabla \theta_o) \cdot d\mathbf{a} = \oint \psi_o \nabla \theta_o \cdot d\mathbf{l} = 2\pi \psi$$

and

$$\Phi_{pol} = - \int_{S_{pol}} \nabla \times (\chi_o \nabla \phi_o) \cdot d\mathbf{a} = - \oint \chi_o \nabla \phi_o \cdot d\mathbf{l} = 2\pi \chi.$$

[The last step follows since $d\mathbf{l}$ and $\nabla \phi_o$ are antiparallel; this is because the contour that encircles S_{pol} (the “doughnut hole” in Fig. 1) must do so in a “positive” direction, i.e., the direction along which an observer would proceed in keeping the enclosed area to his left, and the unit normal pointing “up.”] Thus, we see that the toroidal magnetic flux through a constant- ψ surface is $2\pi\psi$, and the poloidal flux through a constant- χ surface is $2\pi\chi$.

The canonical representation [Eq. (1)] is useful because it allows the differential equations for the magnetic field lines to be expressed as a Hamiltonian system. As it stands, the canonical form of the magnetic field is written in terms of the position vector $\mathbf{r}$. In order to demonstrate the Hamiltonian nature of the field line equations, it is necessary to introduce a transformation to canonical variables. Following Lewis,⁵ we first define a change of variables from an arbitrary set of coordinates that represent the position vector $\mathbf{r}$ to variables (θ, ψ, ϕ) : $\theta = \theta_o(\mathbf{r})$, $\psi = \psi_o(\mathbf{r})$, $\phi = \phi_o(\mathbf{r})$; and we denote the inverse transformation by $\mathbf{r} = \mathbf{R}(\theta, \psi, \phi)$. Next, we form covariant and contravariant basis vectors $\mathbf{e}^j$ and $\mathbf{e}_k$, respectively:^{17,18}

$$\mathbf{e}^1 \equiv \nabla \theta_o(\mathbf{r}), \quad \mathbf{e}^2 \equiv \nabla \psi_o(\mathbf{r}), \quad \mathbf{e}^3 \equiv \nabla \phi_o(\mathbf{r}), \quad (2)$$

and

$$\mathbf{e}_1 \equiv \frac{\partial \mathbf{R}}{\partial \theta}, \quad \mathbf{e}_2 \equiv \frac{\partial \mathbf{R}}{\partial \psi}, \quad \mathbf{e}_3 \equiv \frac{\partial \mathbf{R}}{\partial \phi}. \quad (3)$$

These basis vectors obey the orthogonality relation

$$\mathbf{e}^i \cdot \mathbf{e}_j = \delta_{ij}. \quad (4)$$

Additionally, the two basis sets can be expressed in terms of each other:

$$\mathbf{e}^i = \frac{1}{J} \mathbf{e}_j \times \mathbf{e}_k, \quad \mathbf{e}_i = J \mathbf{e}^j \times \mathbf{e}^k, \quad (5)$$

where

$$J = \mathbf{e}_1 \times \mathbf{e}_2 \cdot \mathbf{e}_3, \quad \frac{1}{J} = \mathbf{e}^1 \times \mathbf{e}^2 \cdot \mathbf{e}^3, \quad (6)$$

and (i, j, k) are a cyclic permutation of $(1, 2, 3)$. If the coordinates that represent $\mathbf{r}$ are Cartesian, then $1/J$ is the Jacobian of the transformation from Cartesian coordinates to (θ, ψ, ϕ) . We denote the function $\chi_o(\mathbf{r})$ expressed in terms of (θ, ψ, ϕ) by $H(\theta, \psi, \phi)$,

$$H(\theta, \psi, \phi) = \chi_o[\mathbf{R}(\theta, \psi, \phi)] \leftrightarrow$$

$$\chi_o(\mathbf{r}) = H[\theta_o(\mathbf{r}), \psi_o(\mathbf{r}), \phi_o(\mathbf{r})].$$

Not surprisingly, H will turn out to be the Hamiltonian.

Now it is a simple matter to show that the magnetic field-line equations are a Hamiltonian system when expressed in terms of the variables (θ, ψ, ϕ) . The equation for the magnetic field lines is

$$\frac{d\mathbf{r}}{ds} = \frac{\mathbf{B}}{B}, \quad (7)$$

where $\mathbf{r}$ is considered to be parameterized in terms of the arclength s along a field line. The magnetic field in Eq. (1) can be expressed in terms of the covariant and contravariant basis vectors and the derivatives of H as

$$\begin{aligned} \mathbf{B} &= \mathbf{e}^2 \times \mathbf{e}^1 + \mathbf{e}^3 \times \left(\frac{\partial H}{\partial \theta} \mathbf{e}^1 + \frac{\partial H}{\partial \psi} \mathbf{e}^2 + \frac{\partial H}{\partial \phi} \mathbf{e}^3 \right) \\ &= \frac{1}{J} \left(-\mathbf{e}_3 + \frac{\partial H}{\partial \theta} \mathbf{e}_2 - \frac{\partial H}{\partial \psi} \mathbf{e}_1 \right). \end{aligned} \quad (8)$$

Here we see that the component of $\mathbf{B}$ along $\mathbf{e}_3$, the “toroidal” component, is $-1/J$. In order that the Jacobian be well defined, the toroidal component of the magnetic field must not vanish in this representation. For this reason, the formalism presented here is useful for application to tokamaks and stellarators, but not to such devices as the reversed-field pinch in which the toroidal component of $\mathbf{B}$ does vanish over some region of space. To handle these cases, Lewis⁵ has given an alternate formulation of the Hamiltonian description. We now consider the coordinates (θ, ψ) along a field line to be parameterized by the coordinate ϕ , so that $d\mathbf{r}/ds$ can be written as

$$\frac{d\mathbf{r}}{ds} = \mathbf{e}_1 \frac{d\theta}{d\phi} \frac{d\phi}{ds} + \mathbf{e}_2 \frac{d\psi}{d\phi} \frac{d\phi}{ds} + \mathbf{e}_3 \frac{d\phi}{ds}. \quad (9)$$

By substituting Eqs. (8) and (9) into Eq. (7) and taking the inner product with the covariant basis vectors, we can eliminate $d\phi/ds$ and find that the equations for the field lines are

$$\frac{d\theta}{d\phi} = \frac{\partial H}{\partial \psi}, \quad \frac{d\psi}{d\phi} = -\frac{\partial H}{\partial \theta}, \quad (10)$$

which have the exact form of Hamilton’s equations; the poloidal angle θ assumes the role of the canonical coordinate, the toroidal flux ψ becomes the canonical momentum, and the toroidal angle ϕ is analogous to time: $\theta \leftrightarrow q$, $\psi \leftrightarrow p$, $\phi \leftrightarrow t$.

An obvious advantage of casting the field-line equations as a Hamiltonian system is that over 130 years of established techniques can be exploited to study magnetic field-line dynamics. These include canonical transformations and Hamilton–Jacobi theory. In addition, there are relatively new approaches, such as Lie series, mappings and symplectic integration methods. These techniques have proved useful in many Hamiltonian treatments of magnetic fields,^{19–21} including a fairly recent investigation of field-line topology in the Joint European Torus.²²

Formulating a Hamiltonian description of magnetic field lines is a difficult task in toroidal geometry, particularly in the case of nonaxisymmetric magnetic fields. Nevertheless, we have been able to formulate such a description for an arbitrary vacuum region outside of which are current-carrying regions. Our approach makes use of a magnetic scalar

potential, Φ , which is a convenient means of representing vacuum magnetic fields; one obtains the magnetic field $\mathbf{B}$ by simply computing the gradient of this scalar: $\mathbf{B} = \nabla \Phi$. The general form of the scalar potential for fields regular at the poloidal axis has already been given.²³ Here, that function $\Phi(\xi, \eta, \varphi)$ has been generalized to account for fields in vacuum regions that surround a possibly nonaxisymmetric current distribution localized about the poloidal axis. Using this generalized magnetic scalar potential in conjunction with Boozer’s canonical representation of the magnetic field, Eq. (1), a set of canonical variables (θ, ψ, ϕ) and a Hamiltonian $H(\theta, \psi, \phi)$ have been found with which any toroidal vacuum magnetic field can be represented. These results are discussed in Sec. III.

The form of the vacuum field-line Hamiltonian presented in Sec. III is quite complicated in the case of nonaxisymmetric fields. Numerical techniques must be invoked to invert expressions which define the transformation to canonical variables, and consequently, it is not possible to give the Hamiltonian H directly in terms of (θ, ψ, ϕ) . For axisymmetric toroidal fields, though, the situation becomes considerably simpler. In that case, one can explicitly write down a Hamiltonian in terms of canonical variables given by

$$\theta = \theta_o(\xi, \eta) = -2 \arctan \left(\sqrt{\frac{1+\xi}{1-\xi}} \tan \frac{\eta}{2} \right), \quad (11)$$

$$\psi = \psi_o(\xi) = B_0 R_0^2 \left(\frac{1}{\sqrt{1-\xi^2}} - 1 \right), \quad (12)$$

$$\phi = \phi_o(\varphi) = \varphi, \quad (13)$$

where B_0 is the value of the toroidal magnetic field on the poloidal axis, R_0 is the radius of the poloidal axis, and ξ , η and φ are toroidal coordinates defined in Appendix A. These functions are easily inverted to give $\xi_o(\psi)$, $\eta_o(\theta, \psi)$ and $\varphi_o(\phi)$. In the “large-aspect-ratio” limit, one can show that $\xi \rightarrow r/R_0$, $\eta \rightarrow \vartheta$ and $\varphi \rightarrow z/R_0$, where (r, ϑ, z) are standard cylindrical coordinates.

With axisymmetry, one can define a gauge in which the covariant representation of the vector potential has two φ -independent components: $\mathbf{A} = A_\eta \nabla \eta + A_\varphi \nabla \varphi$, where $\mathbf{B} = \nabla \times \mathbf{A}$. Here, the term A_η is associated with a magnetic field in the toroidal direction produced by a long straight wire carrying a current I_Z in the negative Z -direction, where Z is defined in Appendix A and illustrated in Fig. 2. In Gaussian cgs units, it has the form

$$A_\eta = -\frac{I_Z R_0}{c} \left[\frac{\ln \left(\frac{1-\xi \cos \eta}{1-\xi} \right)}{1-\cos \eta} + \frac{\ln \left(\frac{1-\xi \cos \eta}{1+\xi} \right)}{1+\cos \eta} \right], \quad (14)$$

where c is the speed of light. The current I_Z is chosen to flow in the negative Z -direction in order to generate a toroidal magnetic field in the negative φ -direction, as required by Eq. (8); see Fig. 3, and note that “ Z ” and “ z ” are distinct and refer to different coordinate systems. The field-line Hamiltonian $H(\theta, \psi)$ is obtained by expressing $-A_\varphi(\xi, \eta)$ in terms of the canonical variables appearing in Eqs. (11) and (12). The above prescription holds for any axisymmetric magnetic

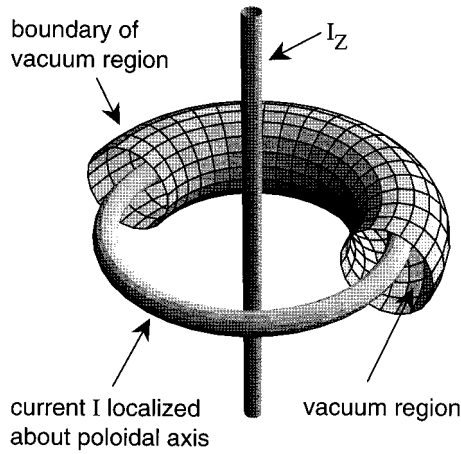


FIG. 2. Toroidal vacuum region. The area of interest is an annular toroidal region between two arbitrary current distributions. The first is localized about the poloidal axis, and the second lies outside the vacuum boundary (e.g., external field coils).

configuration in toroidal geometry, provided that the toroidal component of the magnetic field does not vanish. We shall demonstrate this point in Sec. IV when we consider an axisymmetric example.

III. NONAXISYMMETRIC VACUUM FIELDS

Vacuum magnetic fields obey both $\nabla \cdot \mathbf{B} = 0$ and $\nabla \times \mathbf{B} = 0$. Consequently, a vacuum magnetic field can be obtained from the gradient of a magnetic scalar potential Φ that obeys Laplace's equation: $\nabla^2 \Phi = 0$. In terms of toroidal coordinates, Laplace's equation is separable with periodic, continuous solutions of the form

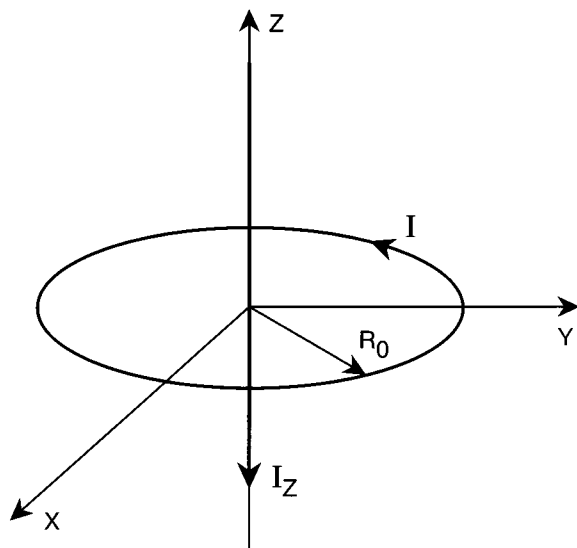


FIG. 3. The current loop. The loop carries a steady current I , has a radius R_0 and lies on the poloidal axis. The straight wire along the Z -axis passes through the center of the loop.

$$\Phi_p(\xi, \eta, \varphi) = \sqrt{1 - \xi \cos \eta} \sum_{m,n} [A_{mn} S_{mn}(\xi) + B_{mn} T_{mn}(\xi)] \times e^{im\eta} e^{in\varphi}, \quad (15)$$

where the complex coefficients A_{mn} and B_{mn} are arbitrary provided that $A_{m,n} = A_{-m,-n}^*$ and $B_{m,n} = B_{-m,-n}^*$, and where

$$S_{mn}(\xi) \equiv \xi^{-1/2} P_{m-1/2}^{[n]}(1/\xi), \quad T_{mn}(\xi) \equiv \xi^{-1/2} Q_{m-1/2}^{[n]}(1/\xi).$$

The functions $P_\nu^\mu(\xi^{-1})$ and $Q_\nu^\mu(\xi^{-1})$ are associated Legendre functions of order μ and degree ν ; they are analytic for $0 < \xi < 1$. The functions $S_{mn}(\xi)$ and $T_{mn}(\xi)$ are singular at $\xi = 0$ and $\xi = 1$, respectively. For convenience, we will adopt the shorthand notation $S_k \equiv S_{k0}(\xi)$, and $T_k \equiv T_{k0}(\xi)$. Note that the potential Φ_p given by Eq. (15) cannot describe a current configuration in which a net current is enclosed by a vacuum region in the poloidal or toroidal direction. That is because Φ_p is periodic in η and φ .

Recall that the Hamiltonian description of field lines used here requires that the toroidal component of the magnetic field not vanish. A convenient means of fulfilling this criterion is to include in our system the vacuum magnetic field produced by a wire lying on the Z -axis and carrying a current I_Z in the negative Z -direction. The magnetic field produced by a line current along the Z -axis is also a good approximation of the field generated by a set of uniformly-spaced toroidal-field coils, which is a common component of toroidal confinement devices. However, the magnetic scalar potential associated with this configuration lies outside the class of functions that can be represented using Eq. (15). To account for such fields, we need to add the secular term $-2I_Z\varphi/c$ to Φ_p . One can easily show that the scalar potential $-2I_Z\varphi/c$ is consistent with the vector potential for a line current along the negative Z -axis given in Eq. (14); both potentials give rise to a toroidal magnetic field $\mathbf{B}_T = -(B_0 R_0 / R) \hat{\varphi}$, where $B_0 \equiv 2I_Z / c R_0$, and R is the distance from the toroidal axis.

In the next subsection, we generalize Eq. (15) further to account for toroidal magnetic fields in vacuum regions that may surround arbitrary current distributions localized in the neighborhood of the poloidal axis.

A. General magnetic scalar potential

Consider the magnetic configuration shown in Fig. 2. The area of interest is an annular vacuum region between two arbitrary current distributions. The first is localized about the poloidal axis, and carries a total current I . This current distribution need not be axisymmetric. The second current distribution is somewhere outside of the outer vacuum boundary. Our objective is to find a general magnetic scalar potential Φ in terms of which any magnetic field in this vacuum region can be described.

The scalar potential Φ_p appearing in Eq. (15) can be used to represent all magnetic fields in the vacuum region produced by *external* currents, e.g., currents residing on the outside surface of a vacuum chamber. However, this scalar potential is incomplete. Although Φ_p contains both the regular and irregular solutions to Laplace's equation, it is missing

a term necessary for describing fields from *internal* sources, i.e., fields generated by currents that are surrounded by the vacuum region. This term can be taken as the potential of a filamentary current loop plus solutions of Laplace's equation that are periodic in η .

In toroidal coordinates, the magnetic scalar potential of a current loop lying on the poloidal axis and carrying a steady current I in the φ -direction (see Fig. 3) is given in Gaussian cgs units by

$$\Phi_l = \frac{2I}{c} \eta + \frac{I}{c} \xi \sum_{k=1}^{\infty} [2S_1 T_k - S_0(T_{k-1} + T_{k+1})] \frac{\sin k \eta}{k}. \quad (16)$$

The derivation of Φ_l appears in Appendix C. It is easily shown that this potential yields the correct Ampèrian loop integral along a path which encircles the poloidal axis once, and also has the correct limiting form for small ξ , i.e., it recovers the result for a straight wire in the limit $\xi \rightarrow 0$. In addition, the Laplacian of Φ_l vanishes except on the poloidal axis, as it should.

The most noteworthy feature of Φ_l is that it is not a single-valued function of position, since the first term on the right side of Eq. (16) is proportional to the poloidal angle η . Panofsky and Phillips²⁴ have cited this multi-valued property as a reason for not using a magnetic scalar potential to describe vacuum magnetic fields in regions surrounding current distributions. However, such an objection does not appear to be warranted. Unlike the vector potential, the magnetic scalar potential is a mathematical construct, and therefore there is no physical restriction limiting it to the class of single-valued functions.

We now demonstrate that Φ_l is the required term which must be added to Φ_p in order to describe fields from internal currents. Referring to Fig. 2, we suppose that the toroidal current density within the vacuum region has a value $\mathbf{j}$, and that this current is localized about the poloidal axis within a constant- ξ contour. For convenience, we label this contour ξ_a . We do not place any requirements on the internal current distribution, except that it is steady-state, and has a total current I . We are interested in the magnetic field $\mathbf{B}$ in the region between ξ_a and the outer boundary of the vacuum region, which we label $\xi = \xi_b$ ($\xi_b > \xi_a$). Since there is no current in this region, the magnetic field is a vacuum field, and thus derivable from a magnetic scalar potential, Φ .

Now imagine a curve C that lies between the ξ_a and ξ_b contours, and denote the surface that C bounds by S . The total current piercing the surface S is the same whether we are considering the arbitrary current distribution $\mathbf{j}$ within the ξ_a -contour, or an infinitesimal current loop with current density $\mathbf{j}_l$ and total current I . Since both configurations carry the same current, we may write

$$I = \int_S \mathbf{j} \cdot d\mathbf{a} = \int_S \mathbf{j}_l \cdot d\mathbf{a}.$$

Ampère's law, $c \nabla \times \mathbf{B} = 4 \pi \mathbf{j}$ and $c \nabla \times \mathbf{B}_l = 4 \pi \mathbf{j}_l$, then gives

$$\int_S \nabla \times \mathbf{B} \cdot d\mathbf{a} = \int_S \nabla \times \mathbf{B}_l \cdot d\mathbf{a}.$$

Finally, using Stokes' theorem we may rewrite these as line integrals over the contour C :

$$\oint_C \nabla \Phi \cdot d\mathbf{l} = \oint_C \nabla \Phi_l \cdot d\mathbf{l},$$

$$\oint_C d(\Phi - \Phi_l) = 0.$$

The last equation implies that $\Phi - \Phi_l$ is a periodic function of η . Such a function must also obey Laplace's equation, and therefore is given by Eq. (15).

Hence, we can say that the general magnetic scalar potential in the vacuum region $\xi_a < \xi < \xi_b$ due to internal ($\xi < \xi_a$) and/or external ($\xi > \xi_b$) current sources will have the form

$$\Phi(\xi, \eta, \varphi) = -\frac{2I_Z \varphi}{c} + \Phi_l(\xi, \eta) + \Phi_p(\xi, \eta, \varphi), \quad (17)$$

where Φ_l and Φ_p are given by Eqs. (16) and (15), respectively, and where we have allowed for a strictly toroidal magnetic field by including the potential of a current I_Z flowing in the negative Z -direction. The value of Φ on the ξ_a and ξ_b contours could in principle be used to determine the two sets of coefficients A_{mn} and B_{mn} .

Using the general form for the magnetic scalar potential given in Eq. (17), we now derive the field-line Hamiltonian in terms of which any toroidal vacuum magnetic field can be described.

B. Field-line Hamiltonian

The first step in deriving the general vacuum field-line Hamiltonian in toroidal geometry is to define the transformation to canonical variables (θ, ψ, ϕ) :

$$\theta = \theta_o(\xi, \eta, \varphi), \quad \psi = \psi_o(\xi, \eta, \varphi), \quad \phi = \phi_o(\xi, \eta, \varphi). \quad (18)$$

We denote the inverse transformation by

$$\xi = \xi_o(\theta, \psi, \phi), \quad \eta = \eta_o(\theta, \psi, \phi), \quad \varphi = \varphi_o(\theta, \psi, \phi). \quad (19)$$

These transformation functions are completely arbitrary at present. In what follows, we determine these functions in part by restricting them to the class of transformations associated with vacuum magnetic fields.

In terms of the canonical variables (θ, ψ, ϕ) , the vacuum condition $\mathbf{B} = \nabla \Phi$ is most easily written in terms of covariant basis vectors, $\mathbf{e}^i$. However, this relation can also be expressed using contravariant basis vectors, $\mathbf{e}_j$:

$$\begin{aligned} \mathbf{B} = \nabla \Phi &= \frac{\partial \Phi}{\partial \theta} \mathbf{e}^1 + \frac{\partial \Phi}{\partial \psi} \mathbf{e}^2 + \frac{\partial \Phi}{\partial \phi} \mathbf{e}^3 \\ &= \left(g^{11} \frac{\partial \Phi}{\partial \theta} + g^{12} \frac{\partial \Phi}{\partial \psi} + g^{13} \frac{\partial \Phi}{\partial \phi} \right) \mathbf{e}_1 \\ &\quad + \left(g^{12} \frac{\partial \Phi}{\partial \theta} + g^{22} \frac{\partial \Phi}{\partial \psi} + g^{23} \frac{\partial \Phi}{\partial \phi} \right) \mathbf{e}_2 \\ &\quad + \left(g^{13} \frac{\partial \Phi}{\partial \theta} + g^{23} \frac{\partial \Phi}{\partial \psi} + g^{33} \frac{\partial \Phi}{\partial \phi} \right) \mathbf{e}_3, \end{aligned} \quad (20)$$

where the g^{ij} 's are inverse metric tensor elements defined in Appendix B. In this form, we may now equate the contravariant components of Eq. (20) with the contravariant components of the general representation of the magnetic field given in Eq. (8) to arrive at the relations

$$\frac{\partial}{\partial \theta} H(\theta, \psi, \phi) = J \left(g^{12} \frac{\partial \Phi}{\partial \theta} + g^{22} \frac{\partial \Phi}{\partial \psi} + g^{23} \frac{\partial \Phi}{\partial \phi} \right), \quad (21)$$

$$-\frac{\partial}{\partial \psi} H(\theta, \psi, \phi) = J \left(g^{11} \frac{\partial \Phi}{\partial \theta} + g^{12} \frac{\partial \Phi}{\partial \psi} + g^{13} \frac{\partial \Phi}{\partial \phi} \right), \quad (22)$$

$$-1 = J \left(g^{13} \frac{\partial \Phi}{\partial \theta} + g^{23} \frac{\partial \Phi}{\partial \psi} + g^{33} \frac{\partial \Phi}{\partial \phi} \right), \quad (23)$$

where $1/J$ is the Jacobian of the transformation from Cartesian to canonical coordinates. Provided that Eq. (23) is satisfied, we are assured that $H_{\psi\theta} = H_{\theta\psi}$, where subscripts denote partial differentiation. This follows from the fact that Φ is a harmonic function. Equality of the mixed partial derivatives is a necessary and sufficient condition for the existence of a function H that satisfies Eqs. (21)–(22).

Using identities such as Eqs. (B20) and (B21) allows Eqs. (21), (22) and (23) to be written as

$$\begin{aligned} \frac{\partial H}{\partial \theta} = \frac{R_0 \xi_o (1 - \xi_o^2)}{1 - \xi_o \cos \eta_o} & \left\{ [\eta_o, \varphi_o]_{\phi, \theta} \frac{\partial \Phi}{\partial \xi} \right. \\ & \left. + \frac{[\varphi_o, \xi_o]_{\phi, \theta}}{\xi_o^2 (1 - \xi_o^2)} \frac{\partial \Phi}{\partial \eta} + \frac{[\xi_o, \eta_o]_{\phi, \theta}}{(1 - \xi_o^2)^2} \frac{\partial \Phi}{\partial \varphi} \right\}, \end{aligned} \quad (24)$$

$$\begin{aligned} -\frac{\partial H}{\partial \psi} = \frac{R_0 \xi_o (1 - \xi_o^2)}{1 - \xi_o \cos \eta_o} & \left\{ [\eta_o, \varphi_o]_{\psi, \phi} \frac{\partial \Phi}{\partial \xi} \right. \\ & \left. + \frac{[\varphi_o, \xi_o]_{\psi, \phi}}{\xi_o^2 (1 - \xi_o^2)} \frac{\partial \Phi}{\partial \eta} + \frac{[\xi_o, \eta_o]_{\psi, \phi}}{(1 - \xi_o^2)^2} \frac{\partial \Phi}{\partial \varphi} \right\}, \end{aligned} \quad (25)$$

$$\begin{aligned} -1 = \frac{R_0 \xi_o (1 - \xi_o^2)}{1 - \xi_o \cos \eta_o} & \left\{ [\eta_o, \varphi_o]_{\theta, \psi} \frac{\partial \Phi}{\partial \xi} \right. \\ & \left. + \frac{[\varphi_o, \xi_o]_{\theta, \psi}}{\xi_o^2 (1 - \xi_o^2)} \frac{\partial \Phi}{\partial \eta} + \frac{[\xi_o, \eta_o]_{\theta, \psi}}{(1 - \xi_o^2)^2} \frac{\partial \Phi}{\partial \varphi} \right\}, \end{aligned} \quad (26)$$

where the square brackets denote Poisson brackets. Equations (24)–(26) comprise a nonlinear set of relations to be solved for the field-line Hamiltonian $H(\theta, \psi, \phi)$ and the three transformation functions $\theta_o(\xi, \eta, \varphi)$, $\psi_o(\xi, \eta, \varphi)$ and $\phi_o(\xi, \eta, \varphi)$. The general solution of Eqs. (24)–(26) is a difficult task, to say the least. In fact, we would have little hope of proceeding if it were not for the fact that there is a certain redundancy built into the canonical description of magnetic field lines. In Sec. II, we saw that the canonical representation of magnetic fields, Eq. (1), involves *four* arbitrary functions of position, whereas a general divergence-free field should involve only *two*. This redundancy translates into the freedom to restrict two of the inverse transformation functions given in Eq. (19). With an appropriate choice of restrictions, Eqs. (24)–(26) can be reduced to a *linear* set of equations that can be solved for the third transformation function and the Hamiltonian.

One class of restrictions that reduces Eqs. (24)–(26) to a linear system of partial differential equations is $\xi = \xi_o(\psi, \phi)$, and $\varphi = \varphi_o(\phi) = \phi$, so that ξ_o is a function of ψ and ϕ only, and $\eta_o(\theta, \psi, \phi)$ is yet to be determined. With these restrictions, Eqs. (24), (25) and (26) become

$$\begin{aligned} \frac{\partial H}{\partial \theta} = -\frac{R_0 \xi_o}{1 - \xi_o \cos \eta_o} \frac{\partial \eta_o}{\partial \theta} & \left\{ (1 - \xi_o^2) \frac{\partial \Phi}{\partial \xi} \right. \\ & \left. - (1 - \xi_o^2)^{-1} \frac{\partial \xi_o}{\partial \phi} \frac{\partial \Phi}{\partial \varphi} \right\}, \end{aligned} \quad (27)$$

$$\begin{aligned} \frac{\partial H}{\partial \psi} = -\frac{R_0 \xi_o (1 - \xi_o^2)}{1 - \xi_o \cos \eta_o} & \left\{ \frac{\partial \eta_o}{\partial \psi} \frac{\partial \Phi}{\partial \xi} \right. \\ & \left. - \frac{\partial \xi_o}{\partial \psi} \frac{1}{(1 - \xi_o^2)^2} \left[\frac{1 - \xi_o^2}{\xi_o^2} \frac{\partial \Phi}{\partial \eta} - \frac{\partial \eta_o}{\partial \phi} \frac{\partial \Phi}{\partial \varphi} \right] \right\}, \end{aligned} \quad (28)$$

$$1 = \frac{R_0 \xi_o}{1 - \xi_o \cos \eta_o} \frac{1}{1 - \xi_o^2} \frac{\partial \xi_o}{\partial \psi} \frac{\partial \eta_o}{\partial \theta} \frac{\partial \Phi}{\partial \varphi}. \quad (29)$$

Since $\eta_o(\theta, \psi, \phi)$ is the only inverse transformation function that depends on the canonical variable θ , we may integrate Eq. (27) to obtain

$$\begin{aligned} H = \frac{R_0 \xi_o}{1 - \xi_o^2} \frac{\partial \xi_o}{\partial \phi} \int_0^{\eta_o} \frac{\Phi_\varphi(\xi_o, y, \varphi_o)}{1 - \xi_o \cos y} dy \\ + R_0 \int_{\xi_a}^{\xi_o} \frac{\Phi_\eta(x, 0, \varphi_o)}{x(1 - x)} dx \\ - R_0 \xi_o (1 - \xi_o^2) \int_0^{\eta_o} \frac{\Phi_\xi(\xi_o, y, \varphi_o)}{1 - \xi_o \cos y} dy, \end{aligned} \quad (30)$$

where ξ_a is a constant between zero and one, and we have used the subscript notation for partial differentiation. Equation (30) is a Hamiltonian for a general three-dimensional toroidal magnetic field in vacuum.

The transformation function $\theta_o(\xi, \eta, \varphi)$ is obtained from Eq. (29) by changing variables from (θ, ψ, ϕ) to (ξ, η, φ) and then integrating Eq. (29) with respect to η . The result is

$$\begin{aligned} \theta_o(\xi, \eta, \varphi) = \frac{R_0 \xi}{1 - \xi^2} \left(\frac{\partial \psi_o}{\partial \xi} \right)^{-1} \sum_{m,n} i n (A_{mn} S_{mn} + B_{mn} T_{mn}) \\ \times \int_0^\eta \frac{e^{in\varphi} e^{imy}}{\sqrt{1 - \xi \cos y}} dy - \frac{4 I_Z R_0}{c} \frac{\xi}{(1 - \xi^2)^{3/2}} \\ \times \left(\frac{\partial \psi_o}{\partial \xi} \right)^{-1} \arctan \left(\sqrt{\frac{1 + \xi}{1 - \xi}} \tan \frac{\eta}{2} \right). \end{aligned} \quad (31)$$

In order that the canonical variable θ be a proper poloidal angle, the function θ_o must change by -2π whenever η changes by 2π . Consequently, the coefficient of the secular term in Eq. (31) must be -1 . This requirement allows us to determine the function ψ_o . Using the fact that

$$\int_0^{2\pi} \frac{\cos m y dy}{\sqrt{1 - \xi \cos y}} = 2 \sqrt{2} T_{m0}(\xi),$$

we find

$$\psi_o(\xi, \varphi) = B_0 R_0^2 \left(\frac{1}{\sqrt{1-\xi^2}} - 1 \right) - \frac{\sqrt{2} R_0}{\pi} \sum_{m,n} i n e^{i n \varphi} \times \int_0^\xi \frac{x T_{m0}(x)}{(1-x^2)} [A_{mn} S_{mn}(x) + B_{mn} T_{mn}(x)] dx. \quad (32)$$

Thus, using the generalized magnetic scalar potential given in Eq. (17), we have found a field-line Hamiltonian $H(\theta, \psi, \phi)$ and associated canonical variables (θ, ψ, ϕ) in terms of which any toroidal vacuum magnetic field can be described. This includes nonaxisymmetric fields in vacuum regions surrounding arbitrary current distributions. Unfortunately, though, the Hamiltonian appearing in Eq. (30) cannot be expressed explicitly in terms of the variables (θ, ψ, ϕ) . To do so would require the determination of the transformation functions $\xi_o(\theta, \psi, \phi)$ and $\eta_o(\theta, \psi, \phi)$ through the inversion of Eqs. (31) and (32). Because of the multi-valued nature of these functions, this would be an arduous task even with sophisticated numerical techniques.

The difficulties associated with inverting transformation functions disappear in axisymmetric systems. In this case, $n=0$ in the summations of Eqs. (31) and (32), which are then identical to Eqs. (11) and (12), respectively. These equations are easily inverted to give $\xi_o(\psi)$ and $\eta_o(\theta, \psi)$. A careful inspection of Eq. (11) shows that θ has the correct secular behavior, i.e., it changes by -2π whenever η changes by 2π . Although the toroidal canonical variables defined in Eqs. (11) and (12) were derived for vacuum magnetic fields, it turns out that they are not restricted to this class; Eqs. (11) and (12) apply to any axisymmetric field configuration in toroidal geometry provided the toroidal field is everywhere finite. In the next section, we demonstrate this fact, and use these variables to derive the field-line Hamiltonian for axisymmetric systems.

IV. AXISYMMETRIC SYSTEMS

In this section, a field-line Hamiltonian for an arbitrary axisymmetric magnetic-field configuration is determined. The field-line Hamiltonian $H(\theta, \psi)$ is obtained by first deriving the covariant φ -component of the vector potential, $A_\varphi(\xi, \eta)$, and then expressing the negative of this quantity in terms of the canonical variables defined in Eqs. (11) and (12).

For axisymmetric vacuum fields, the magnetic field-line Hamiltonian may be found by using a φ -independent scalar potential, $\Phi(\xi, \eta)$, in Eq. (30), which leads to

$$H(\theta, \psi) = -R_0 \xi_o (1 - \xi_o^2) \int_0^{\eta_o} \frac{\Phi_\xi(\xi_o, y)}{1 - \xi_o \cos y} dy + R_0 \int_{\xi_a}^{\xi_o} \frac{\Phi_\eta(x, 0)}{x(1-x)} dx.$$

These integrals can be evaluated analytically, with the result

$$H(\theta, \psi) = \frac{B_0 R_0^2 \sqrt{1 - \xi_o^2}}{\sqrt{1 - \xi_o \cos \eta_o}} \times \sum_{m=-\infty}^{\infty} [a_m S_{m-1}(\xi_o) + b_m T_{m-1}(\xi_o)] e^{i m \eta_o}, \quad (33)$$

where we have omitted an unimportant constant term, and the subscript “ o ” denotes dependence on the canonical variables, $\xi_o \equiv \xi_o(\psi)$ and $\eta_o \equiv \eta_o(\theta, \psi)$, and where $a_m = a_{-m}^*$ and $b_m = b_{-m}^*$ are arbitrary, complex dimensionless constants. Equation (33) is the general form of the field-line Hamiltonian for axisymmetric vacuum magnetic fields in toroidal geometry. The current loop corresponds to the a_0 term.

It is straightforward to demonstrate that Eqs. (11) and (12) also define canonical variables that can be used to construct a field-line Hamiltonian for any axisymmetric magnetic-field configuration in toroidal geometry. To show this, we first write down the equations for the magnetic field lines in (ξ, η, φ) coordinates in terms of the contravariant components of the magnetic field:

$$\frac{d\xi}{d\varphi} = \frac{B^\xi}{B^\varphi}, \quad \frac{d\eta}{d\varphi} = \frac{B^\eta}{B^\varphi}.$$

The contravariant components are related to the covariant components of the vector potential A_i according to

$$\mathcal{J} B^\xi = \frac{\partial A_\varphi}{\partial \eta} - \frac{\partial A_\eta}{\partial \varphi},$$

$$\mathcal{J} B^\eta = \frac{\partial A_\xi}{\partial \varphi} - \frac{\partial A_\varphi}{\partial \xi},$$

$$\mathcal{J} B^\varphi = \frac{\partial A_\eta}{\partial \xi} - \frac{\partial A_\xi}{\partial \eta},$$

where $1/\mathcal{J}$ is the Jacobian of the transformation from Cartesian to toroidal coordinates. For any axisymmetric toroidal configuration with a toroidal magnetic field $B_T \sim 1/R$, we can define a gauge in which $A_\xi = 0$, and A_η is given by Eq. (14). The partial derivative of A_η with respect to ξ is

$$\frac{\partial A_\eta}{\partial \xi} = \frac{B_0 R_0^2 \xi}{(1 - \xi^2)(1 - \xi \cos \eta)}.$$

The equations for the magnetic field lines then become

$$\frac{d\xi}{d\varphi} = + \frac{1}{B_0 R_0^2 \xi} (1 - \xi^2)(1 - \xi \cos \eta) \frac{\partial A_\varphi}{\partial \eta}, \quad (34)$$

$$\frac{d\eta}{d\varphi} = - \frac{1}{B_0 R_0^2 \xi} (1 - \xi^2)(1 - \xi \cos \eta) \frac{\partial A_\varphi}{\partial \xi}, \quad (35)$$

which are obviously not in Hamiltonian form. However, by using Eqs. (11) and (12) to change variables from (ξ, η) to (θ, ψ) , we can effectively “absorb” the $\xi^{-1}(1 - \xi^2) \times (1 - \xi \cos \eta)/B_0 R_0^2$ term in front of the partial derivatives in Eqs. (34) and (35), and recast the magnetic field-line equations as a Hamiltonian system. The resulting “equations of motion” are

$$\frac{d\theta}{d\varphi} = + \frac{\partial H}{\partial \psi}, \quad \frac{d\psi}{d\varphi} = - \frac{\partial H}{\partial \theta},$$

which are Hamilton's equations with $H(\theta, \psi) \equiv -A_\varphi[\xi(\psi), \eta(\theta, \psi)]$. There is no reason to restrict A_φ to the class of potentials associated with vacuum magnetic fields. The determination of A_φ will in general depend on the current distribution of a particular system.

As an example, we now turn to an analysis of a filamentary current loop. For this system, we derive an explicit expression for A_φ , which may then be transformed to a field-line Hamiltonian $H(\theta, \psi)$ by using Eqs. (11) and (12).

A. Example: The current loop

Consider a circular, filamentary current loop. The loop carries a steady current I , and has a radius R_0 . It lies in the X - Y plane on the poloidal axis with the Z -axis passing through its center (see Fig. 3). In addition, we imagine that there is a straight wire carrying a current I_Z along the negative- Z direction.

The magnetic field generated by the straight wire and the current loop is axisymmetric; all components of the magnetic field are independent of the toroidal angle φ . In this case, we can define the gauge $A_\xi=0$, so that the covariant representation of the vector potential has only two components, A_η and A_φ , where A_η is given by Eq. (14). The other nonzero component, A_φ , is derived in Appendix C. When expressed in terms of canonical variables (θ, ψ) , $-A_\varphi$ is the field-line Hamiltonian for the current loop. We find

$$H(\theta, \psi) = \frac{B_0 R_0^2}{\delta_I} \left[\hat{\psi} + \sqrt{\hat{\psi}^2 - 1} \cos \theta \right]^{1/2} Q_{1/2}(\hat{\psi})$$

or

$$H(\theta, \psi) = \frac{\sqrt{\pi} B_0 R_0^2}{2 \delta_I} Q_{1/2}(\hat{\psi}) \times \sum_{k=0}^{\infty} \frac{\epsilon_k}{\Gamma\left(k + \frac{3}{2}\right)} P_{1/2}^k(\hat{\psi}) \cos k\theta, \quad (36)$$

where $\hat{\psi} \equiv 1 + \psi/B_0 R_0^2$, $\delta_I \equiv I_Z/I$, $Q_{1/2}$ and $P_{1/2}^k$ are Legendre functions, and ϵ_k is the Neumann number:

$$\epsilon_k = \begin{cases} 1, & \text{if } k=0, \\ 2, & \text{otherwise.} \end{cases}$$

The average of the Hamiltonian over θ , $\langle H \rangle$, which is the first term in Eq. (36), is given by

$$\langle H \rangle \equiv \frac{1}{2\pi} \int_0^{2\pi} H(\theta, \psi) d\theta = \frac{B_0 R_0^2}{\delta_I} Q_{1/2}(\hat{\psi}) P_{1/2}(\hat{\psi}). \quad (37)$$

Using this expression, we can estimate the safety factor q for this system near the poloidal axis. Recall that the safety factor is inversely proportional to the rotational transform, ι , which is the η -angle covered by a given field line in one complete toroidal revolution:

$$\frac{1}{q} = \frac{\iota}{2\pi} = \frac{1}{2\pi} \int_0^{2\pi} \frac{d\eta}{d\varphi} d\varphi. \quad (38)$$

Since the first term in the series for H is dominant near $\psi=0$ (which corresponds to $\hat{\psi}=1$), the safety factor near the poloidal axis can be approximated by expanding $\langle H \rangle$ in a Taylor series about $\psi=0$. Near the poloidal axis, we also have that $\psi \rightarrow B_0 R_0^2 \xi^2/2 \rightarrow B_0 r^2/2$, and $\theta \rightarrow -\eta \rightarrow -\vartheta$. To lowest order, Eq. (38) becomes

$$\frac{1}{q} = -\frac{\partial H}{\partial \psi} \sim \frac{1}{\delta_I} \frac{R_0^2}{r^2}.$$

Thus, near the poloidal axis the safety factor varies quadratically with the minor radius.

To include a vertical magnetic field in this description, one simply needs to add to H a term

$$\frac{B_v R_0^2}{2} \left[\hat{\psi} + \sqrt{\hat{\psi}^2 - 1} \cos \theta \right]^2,$$

where B_v is the strength of the vertical field. Freis *et al.*²⁵ considered such a model as the basis of a Hamiltonian description of an axisymmetric levitron device. Their analysis, however, differs from the present one in that only first-order toroidal effects were incorporated into the field-line Hamiltonian.

V. DISCUSSION AND CONCLUSIONS

In this paper, we have reported some results associated with the Hamiltonian formulation of toroidal magnetic fields. We have presented a general magnetic scalar potential in terms of which any toroidal vacuum magnetic field can be written, even those surrounding a toroidal current distribution localized about the poloidal axis. Using this potential, we derived an expression for a general (nonaxisymmetric) field-line Hamiltonian which governs the flow of vacuum magnetic field lines in toroidal geometry.

The Hamiltonian for a nonaxisymmetric, vacuum magnetic field can be written in the form

$$H(\theta, \psi, \phi) = H_0(\theta, \psi) + \varepsilon H_1(\theta, \psi, \phi),$$

where H_0 describes the axisymmetric component of the field and ε is proportional to the strength of the nonaxisymmetric component. In this form, the topological properties of non-axisymmetric magnetic fields can be investigated using a perturbation theory developed by Lewis, Bates and Finn for constructing invariants of Hamiltonians order by order in powers of ε .¹⁰ Recently,⁹ approximate field-line invariants obtained with this procedure were used to examine the topological properties of selected vacuum and MHD-equilibrium magnetic-field configurations that are slightly perturbed from axisymmetry. Cases were examined in toroidal geometry and in a periodic cylindrical configuration. Excellent agreement between Poincaré maps and the level-curves of the first-order invariant was found for small perturbations. A means of circumventing the "small divisor problem" in some cases was identified and implemented with outstanding results. The examples indicate that this perturbation method can have valuable consequences in future investigations of magnetic field-line topology. A discussion of these results is planned for a future publication.²⁶

ACKNOWLEDGMENTS

The authors acknowledge useful discussions with A. Boozer and J. Tataronis. In addition, we thank S. Geimer and P. Kostelec for their assistance in producing the figures.

This work was supported by a Gordon F. Hull Fellowship at Dartmouth College.

APPENDIX A: TOROIDAL GEOMETRY

The toroidal coordinates (ξ, η, φ) used here are the same as discussed by Cary.²⁷ They are shown in Fig. 4, and obey the transformation

$$\begin{aligned} X &= \frac{R_0 \sqrt{1-\xi^2} \cos \varphi}{1-\xi \cos \eta}, \\ Y &= \frac{R_0 \sqrt{1-\xi^2} \sin \varphi}{1-\xi \cos \eta}, \\ Z &= -\frac{R_0 \xi \sin \eta}{1-\xi \cos \eta}, \end{aligned} \quad (\text{A1})$$

where X , Y and Z are ordinary Cartesian coordinates, $R^2 = X^2 + Y^2 + Z^2$ and $R_0 \equiv \sqrt{r_0^2 - a^2}$ is the location of the poloidal axis. The parameters r_0 and a are the major and minor radii, respectively, of a toroidal system.

The transformation from Cartesian coordinates to toroidal coordinates is given by

$$\xi = \sqrt{1-\rho^2}, \quad \tan \eta = -\frac{\rho Z}{R-\rho R_0}, \quad (\text{A2})$$

where $\rho \equiv 2RR_0/(R^2 + Z^2 + R^2)$. The variable ξ is similar to a minor radius, η is a poloidal angle and φ is the usual toroidal angle. Surfaces of constant ξ are circular tori (see Fig. 4). The tori have major radii $R_0/\sqrt{1-\xi^2}$ and minor radii $R_0\xi/\sqrt{1-\xi^2}$. Thus, their inverse aspect ratios are given by ξ . Note that ξ is bounded between zero and one: $0 \leq \xi \leq 1$. Also note that the variable η is truly a poloidal angle: one complete circuit around the poloidal axis corresponds to a change of η by 2π .

The toroidal coordinate system is orthogonal with the scale factors

$$\begin{aligned} h_\xi &= \frac{R_0}{\sqrt{1-\xi^2}(1-\xi \cos \eta)}, \quad h_\eta = \frac{R_0 \xi}{1-\xi \cos \eta}, \\ h_\varphi &= \frac{R_0 \sqrt{1-\xi^2}}{1-\xi \cos \eta}, \end{aligned} \quad (\text{A3})$$

and a Jacobian

$$\mathcal{J} = \frac{R_0^3 \xi}{(1-\xi \cos \eta)^3}. \quad (\text{A4})$$

The triad $(\hat{\xi}, \hat{\eta}, \hat{\varphi})$ forms a right-handed, orthonormal basis set: $\hat{\xi} \times \hat{\eta} = \hat{\varphi}$, $\hat{\eta} \times \hat{\varphi} = \hat{\xi}$, $\hat{\varphi} \times \hat{\xi} = \hat{\eta}$. Unit vectors $\hat{\xi}$, $\hat{\eta}$ and $\hat{\varphi}$ in the directions of increasing ξ , η and φ , respectively, are given in terms of unit vectors $\hat{x}$, $\hat{y}$ and $\hat{z}$ in the directions of increasing X , Y , and Z , respectively, by

$$\hat{n} \equiv \hat{x} \cos \varphi + \hat{y} \sin \varphi, \quad (\text{A5})$$

$$\hat{\varphi} = -\hat{x} \sin \varphi + \hat{y} \cos \varphi, \quad (\text{A6})$$

$$\hat{\xi} = \hat{n} \frac{\cos \eta - \xi}{1-\xi \cos \eta} - \hat{z} \frac{\sqrt{1-\xi^2} \sin \eta}{1-\xi \cos \eta}, \quad (\text{A7})$$

$$\hat{\eta} = -\hat{n} \frac{\sqrt{1-\xi^2} \sin \eta}{1-\xi \cos \eta} - \hat{z} \frac{\cos \eta - \xi}{1-\xi \cos \eta}. \quad (\text{A8})$$

APPENDIX B: GENERAL CANONICAL VARIABLES

In this appendix, we provide the relations necessary for carrying out a change of variables from the toroidal coordinates (ξ, η, φ) to the canonical variables (θ, ψ, ϕ) . We give general (unrestricted) expressions for the metric tensor elements, and the covariant and contravariant basis vectors. Following the notation of Sec. III, we first define the transformation to variables (θ, ψ, ϕ) by

$$\theta = \theta_o(\xi, \eta, \varphi), \quad \psi = \psi_o(\xi, \eta, \varphi), \quad \phi = \phi_o(\xi, \eta, \varphi),$$

and denote the inverse transformation by

$$\xi = \xi_o(\theta, \psi, \phi), \quad \eta = \eta_o(\theta, \psi, \phi), \quad \varphi = \varphi_o(\theta, \psi, \phi).$$

In what follows, the subscript notation for partial derivatives is used.

The contravariant basis vectors are

$$\begin{aligned} \mathbf{e}_1 &= \frac{R_0 \xi_{o\theta}}{\sqrt{1-\xi^2}} (1-\xi \cos \eta) \hat{\xi} + \frac{R_0 \xi \eta_{o\theta}}{1-\xi \cos \eta} \hat{\eta} \\ &\quad + \frac{R_0 \sqrt{1-\xi^2} \varphi_{o\theta}}{1-\xi \cos \eta} \hat{\varphi}, \end{aligned} \quad (\text{B1})$$

$$\begin{aligned} \mathbf{e}_2 &= \frac{R_0 \xi_{o\psi}}{\sqrt{1-\xi^2}} (1-\xi \cos \eta) \hat{\xi} + \frac{R_0 \xi \eta_{o\psi}}{1-\xi \cos \eta} \hat{\eta} \\ &\quad + \frac{R_0 \sqrt{1-\xi^2} \varphi_{o\psi}}{1-\xi \cos \eta} \hat{\varphi}, \end{aligned} \quad (\text{B2})$$

$$\begin{aligned} \mathbf{e}_3 &= \frac{R_0 \xi_{o\phi}}{\sqrt{1-\xi^2}} (1-\xi \cos \eta) \hat{\xi} + \frac{R_0 \xi \eta_{o\phi}}{1-\xi \cos \eta} \hat{\eta} \\ &\quad + \frac{R_0 \sqrt{1-\xi^2} \varphi_{o\phi}}{1-\xi \cos \eta} \hat{\varphi}. \end{aligned} \quad (\text{B3})$$

The elements of the metric tensor are

$$\begin{aligned} g_{11} &= \frac{R_0^2}{(1-\xi_o \cos \eta_o)^2} [(1-\xi^2)^{-1} \xi_{o\theta}^2 + \xi^2 \eta_{o\theta}^2 \\ &\quad + (1-\xi^2) \varphi_{o\theta}^2], \end{aligned} \quad (\text{B4})$$

$$\begin{aligned} g_{22} &= \frac{R_0^2}{(1-\xi_o \cos \eta_o)^2} [(1-\xi^2)^{-1} \xi_{o\psi}^2 + \xi^2 \eta_{o\psi}^2 \\ &\quad + (1-\xi^2) \varphi_{o\psi}^2], \end{aligned} \quad (\text{B5})$$

$$\begin{aligned} g_{33} &= \frac{R_0^2}{(1-\xi_o \cos \eta_o)^2} [(1-\xi^2)^{-1} \xi_{o\phi}^2 + \xi^2 \eta_{o\phi}^2 \\ &\quad + (1-\xi^2) \varphi_{o\phi}^2], \end{aligned} \quad (\text{B6})$$

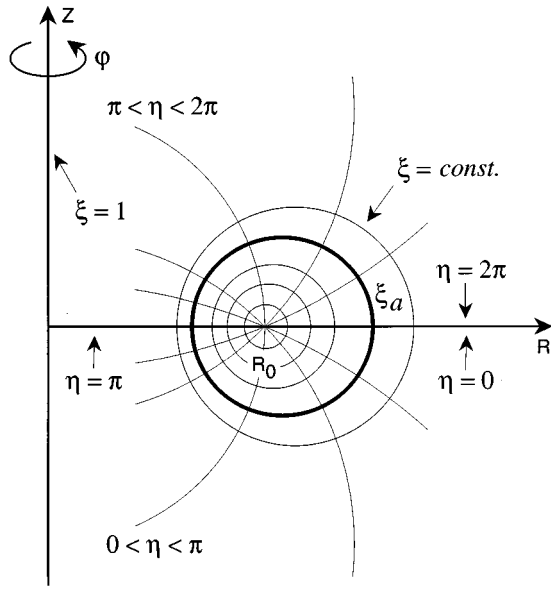


FIG. 4. The toroidal coordinate system (ξ, η, φ) .

$$g_{12} = \frac{R_0^2}{(1 - \xi_o \cos \eta_o)^2} \left[(1 - \xi^2)^{-1} \xi_o \theta \xi_o \psi + \xi^2 \eta_o \theta \eta_o \psi + (1 - \xi^2) \varphi_o \theta \varphi_o \psi \right], \quad (\text{B7})$$

$$g_{13} = \frac{R_0^2}{(1 - \xi_o \cos \eta_o)^2} \left[(1 - \xi^2)^{-1} \xi_o \theta \xi_o \phi + \xi^2 \eta_o \theta \eta_o \phi + (1 - \xi^2) \varphi_o \theta \varphi_o \phi \right], \quad (\text{B8})$$

$$g_{23} = \frac{R_0^2}{(1 - \xi_o \cos \eta_o)^2} \left[(1 - \xi^2)^{-1} \xi_o \psi \xi_o \phi + \xi^2 \eta_o \psi \eta_o \phi + (1 - \xi^2) \varphi_o \psi \varphi_o \phi \right]. \quad (\text{B9})$$

Note the symmetry property $g_{ij} = g_{ji}$. The Jacobian is given by

$$J = \frac{R_0^3 \xi_o}{(1 - \xi_o \cos \eta_o)^3} \left\{ \xi_o \phi [\eta_o, \varphi_o]_{\theta, \psi} + \eta_o \phi [\varphi_o, \xi_o]_{\theta, \psi} + \varphi_o \phi [\xi_o, \eta_o]_{\theta, \psi} \right\}, \quad (\text{B10})$$

where the symbol $[A, B]_{\alpha, \beta}$ denotes the Poisson Bracket of A and B with respect to the variables α and β . The covariant basis vectors are

$$\mathbf{e}^1 = \frac{R_0^2 \xi_o \sqrt{1 - \xi_o^2}}{J(1 - \xi_o \cos \eta_o)^2} \left\{ [\eta_o, \varphi_o]_{\psi, \phi} \hat{\xi} + \frac{[\varphi_o, \xi_o]_{\psi, \phi}}{\xi_o \sqrt{1 - \xi_o^2}} \hat{\eta} + \frac{[\xi_o, \eta_o]_{\psi, \phi}}{(1 - \xi_o^2)} \hat{\varphi} \right\}, \quad (\text{B11})$$

$$\mathbf{e}^2 = \frac{R_0^2 \xi_o \sqrt{1 - \xi_o^2}}{J(1 - \xi_o \cos \eta_o)^2} \left\{ [\eta_o, \varphi_o]_{\phi, \theta} \hat{\xi} + \frac{[\varphi_o, \xi_o]_{\phi, \theta}}{\xi_o \sqrt{1 - \xi_o^2}} \hat{\eta} + \frac{[\xi_o, \eta_o]_{\phi, \theta}}{(1 - \xi_o^2)} \hat{\varphi} \right\}, \quad (\text{B12})$$

$$\mathbf{e}^3 = \frac{R_0^2 \xi_o \sqrt{1 - \xi_o^2}}{J(1 - \xi_o \cos \eta_o)^2} \left\{ [\eta_o, \varphi_o]_{\theta, \psi} \hat{\xi} + \frac{[\varphi_o, \xi_o]_{\theta, \psi}}{\xi_o \sqrt{1 - \xi_o^2}} \hat{\eta} + \frac{[\xi_o, \eta_o]_{\theta, \psi}}{(1 - \xi_o^2)} \hat{\varphi} \right\}. \quad (\text{B13})$$

The inverse metric tensor elements ($g^{ij} = g^{ji}$) are

$$g^{11} = \frac{R_0^4 \xi_o^2 (1 - \xi_o^2)}{J^2 (1 - \xi_o \cos \eta_o)^4} \left\{ [\eta_o, \varphi_o]_{\psi, \phi}^2 + \frac{[\varphi_o, \xi_o]_{\psi, \phi}^2}{\xi_o^2 (1 - \xi_o^2)} + \frac{[\xi_o, \eta_o]_{\psi, \phi}^2}{(1 - \xi_o^2)^2} \right\}, \quad (\text{B14})$$

$$g^{22} = \frac{R_0^4 \xi_o^2 (1 - \xi_o^2)}{J^2 (1 - \xi_o \cos \eta_o)^4} \left\{ [\eta_o, \varphi_o]_{\phi, \theta}^2 + \frac{[\varphi_o, \xi_o]_{\phi, \theta}^2}{\xi_o^2 (1 - \xi_o^2)} + \frac{[\xi_o, \eta_o]_{\phi, \theta}^2}{(1 - \xi_o^2)^2} \right\}, \quad (\text{B15})$$

$$g^{33} = \frac{R_0^4 \xi_o^2 (1 - \xi_o^2)}{J^2 (1 - \xi_o \cos \eta_o)^4} \left\{ [\eta_o, \varphi_o]_{\theta, \psi}^2 + \frac{[\varphi_o, \xi_o]_{\theta, \psi}^2}{\xi_o^2 (1 - \xi_o^2)} + \frac{[\xi_o, \eta_o]_{\theta, \psi}^2}{(1 - \xi_o^2)^2} \right\}, \quad (\text{B16})$$

$$g^{12} = \frac{R_0^4 \xi_o^2 (1 - \xi_o^2)}{J^2 (1 - \xi_o \cos \eta_o)^4} \left\{ [\eta_o, \varphi_o]_{\psi, \phi} [\eta_o, \varphi_o]_{\phi, \theta} + \frac{[\varphi_o, \xi_o]_{\psi, \phi} [\varphi_o, \xi_o]_{\phi, \theta}}{\xi_o^2 (1 - \xi_o^2)} + \frac{[\xi_o, \eta_o]_{\psi, \phi} [\xi_o, \eta_o]_{\phi, \theta}}{(1 - \xi_o^2)^2} \right\}, \quad (\text{B17})$$

$$g^{13} = \frac{R_0^4 \xi_o^2 (1 - \xi_o^2)}{J^2 (1 - \xi_o \cos \eta_o)^4} \left\{ [\eta_o, \varphi_o]_{\psi, \phi} [\eta_o, \varphi_o]_{\theta, \psi} + \frac{[\varphi_o, \xi_o]_{\psi, \phi} [\varphi_o, \xi_o]_{\theta, \psi}}{\xi_o^2 (1 - \xi_o^2)} + \frac{[\xi_o, \eta_o]_{\psi, \phi} [\xi_o, \eta_o]_{\theta, \psi}}{(1 - \xi_o^2)^2} \right\}, \quad (\text{B18})$$

$$g^{23} = \frac{R_0^4 \xi_o^2 (1 - \xi_o^2)}{J^2 (1 - \xi_o \cos \eta_o)^4} \left\{ [\eta_o, \varphi_o]_{\phi, \theta} [\eta_o, \varphi_o]_{\theta, \psi} + \frac{[\varphi_o, \xi_o]_{\phi, \theta} [\varphi_o, \xi_o]_{\theta, \psi}}{\xi_o^2 (1 - \xi_o^2)} + \frac{[\xi_o, \eta_o]_{\phi, \theta} [\xi_o, \eta_o]_{\theta, \psi}}{(1 - \xi_o^2)^2} \right\}. \quad (\text{B19})$$

Additionally, there are the relations

$$\begin{aligned}
& \xi_{o\phi} [\eta_o, \varphi_o]_{\theta, \psi} + \eta_{o\phi} [\varphi_o, \xi_o]_{\theta, \psi} + \varphi_{o\phi} [\xi_o, \eta_o]_{\theta, \psi} \\
& = \xi_{o\theta} [\eta_o, \varphi_o]_{\psi, \phi} + \eta_{o\theta} [\varphi_o, \xi_o]_{\psi, \phi} + \varphi_{o\theta} [\xi_o, \eta_o]_{\psi, \phi}, \\
& = \xi_{o\psi} [\eta_o, \varphi_o]_{\phi, \theta} + \eta_{o\psi} [\varphi_o, \xi_o]_{\phi, \theta} + \varphi_{o\psi} [\xi_o, \eta_o]_{\phi, \theta},
\end{aligned} \tag{B20}$$

and

$$\varphi_{o\theta} [\eta_o, \varphi_o]_{\psi, \phi} + \varphi_{o\psi} [\eta_o, \varphi_o]_{\phi, \theta} + \varphi_{o\phi} [\eta_o, \varphi_o]_{\theta, \psi} = 0. \tag{B21}$$

APPENDIX C: MAGNETIC POTENTIALS FOR A CURRENT LOOP

Here, we derive both the vector and scalar potentials for an axisymmetric current loop in toroidal coordinates. The loop lies on the poloidal axis, has a radius R_0 and carries a steady current I in the φ -direction (see Fig. 3).

Since the system is axisymmetric, the only nonzero component of the vector potential is A_φ : $\mathbf{A}_l = A_\varphi \nabla \varphi$. The vector potential is obtained from the current density $\mathbf{j}$ according to²⁸

$$\mathbf{A}(\mathbf{x}) = \frac{1}{c} \int \frac{\mathbf{j}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x', \tag{C1}$$

where

$$\frac{1}{|\mathbf{x} - \mathbf{x}'|} = \frac{\sqrt{1 - \xi \cos \eta} \sqrt{1 - \xi' \cos \eta'}}{\sqrt{2} R_0 [1 - \xi \xi' \cos(\eta - \eta') - \sqrt{1 - \xi^2} \sqrt{1 - \xi'^2} \cos(\varphi - \varphi')]^{1/2}}.$$

For the circular loop, the current density $\mathbf{j}_l$ has only a φ -component: $\mathbf{j}_l = j_\varphi \hat{\boldsymbol{\varphi}}$. In toroidal coordinates (ξ, η) , it is given by

$$j_\varphi(\xi, \eta) = \frac{I}{2\pi} \delta(\xi) \frac{\sqrt{1 - \xi^2} (1 - \xi \cos \eta)^2}{R_0^2 \xi}. \tag{C2}$$

Since $\xi=0$ corresponds to $R=R_0$, the delta function restricts the current flow to the poloidal axis. In Cartesian coordinates, the current density $\mathbf{j}_l$ is written as

$$\mathbf{j}_l = -j_\varphi \sin \varphi \hat{\mathbf{x}} + j_\varphi \cos \varphi \hat{\mathbf{y}}.$$

Because of the symmetry in this problem, any convenient value of the toroidal angle may be chosen when evaluating the integral in Eq. (C1). For convenience, we choose $\varphi=0$. Furthermore, only the Y -component of $\mathbf{j}_l$ contributes to $\mathbf{A}_l$, so we have

$$\begin{aligned}
A_\varphi &= \frac{IR_0^2}{2\pi c} \frac{\sqrt{1 - \xi^2}}{1 - \xi \cos \eta} \int \delta(\xi') \frac{\cos \varphi' \sqrt{1 - \xi'^2}}{(1 - \xi' \cos \eta') |\mathbf{x} - \mathbf{x}'|} \\
&\quad \times d\xi' d\eta' d\varphi'.
\end{aligned}$$

Integration over the ξ' and η' variables yields

$$\begin{aligned}
A_\varphi &= \frac{2IR_0}{\sqrt{2}c} \frac{\sqrt{1 - \xi^2}}{\sqrt{1 - \xi \cos \eta}} \int_0^\pi \frac{\cos \varphi' d\varphi'}{[1 - \sqrt{1 - \xi^2} \cos \varphi']^{1/2}} \\
&= \frac{2IR_0}{c} \frac{(1 - \xi^2)^{1/4}}{\sqrt{1 - \xi \cos \eta}} Q_{1/2} \left(\frac{1}{\sqrt{1 - \xi^2}} \right) \\
&= -\frac{4\pi IR_0}{\sqrt{2}c} \frac{\sqrt{1 - \xi^2}}{\sqrt{1 - \xi \cos \eta}} S_{01}(\xi).
\end{aligned} \tag{C3}$$

Using the vector potential given in Eq. (C3), we now derive the magnetic scalar potential for a current loop. The first step is to rewrite Eq. (C3) in the form

$$A_\varphi = -\frac{2\pi IR_0}{\sqrt{2}c} \frac{\xi S_{10}(\xi) - S_{00}(\xi)}{\sqrt{1 - \xi \cos \eta}}.$$

Since $\mathbf{B}_l = \nabla \Phi_l = \nabla \times \mathbf{A}_l$, we have that

$$\frac{\partial \Phi_l}{\partial \xi} = \frac{h_\xi}{h_\eta h_\varphi} \frac{\partial A_\varphi}{\partial \eta} = \frac{2\pi I}{\sqrt{2}c} \frac{S_{01}(\xi) \sin \eta}{\sqrt{1 - \xi^2} \sqrt{1 - \xi \cos \eta}}, \tag{C4}$$

$$\frac{\partial \Phi_l}{\partial \eta} = -\frac{h_\eta}{h_\xi h_\varphi} \frac{\partial A_\varphi}{\partial \xi} = \frac{\pi I}{\sqrt{2}c} \frac{\xi S_{10}(\xi) - \xi S_{00}(\xi) \cos \eta}{\sqrt{1 - \xi \cos \eta}}. \tag{C5}$$

Then, using the expansion

$$(1 - \xi \cos \eta)^{-1/2} = \frac{\sqrt{2}}{\pi} \sum_{k=0}^{\infty} \epsilon_k T_{k0}(\xi) \cos k\eta,$$

and the fact that $S_{10}T_{00} - S_{00}T_{10} = 2/\xi$, we may integrate Eq. (C5) to obtain

$$\Phi_l = \frac{2I}{c} \eta + \frac{I}{c} \xi \sum_{k=1}^{\infty} [2S_1 T_k - S_0 (T_{k-1} + T_{k+1})] \frac{\sin k\eta}{k}. \tag{C6}$$

The magnetic scalar potential for a current loop given in Eq. (C6) has the correct secular behavior, and reduces to the result for a straight wire near the poloidal axis ($\xi \rightarrow 0$). The three-term recurrence relation (recall $T_k \equiv T_{k0}$),

$$-2kT_k + \xi \left(k - \frac{1}{2} \right) T_{k-1} + \xi \left(k + \frac{1}{2} \right) T_{k+1} = 0, \tag{C7}$$

the formula for derivatives,

$$\frac{dT_k}{d\xi} = \frac{(\xi/2 - k/\xi) T_k - (k - 1/2) T_{k-1}}{1 - \xi^2}, \tag{C8}$$

and the symmetry property $T_{-k} = T_k$ can be used to verify that the partial derivative of Eq. (C6) with respect to ξ gives

Eq. (C4). Equations (C7) and (C8) hold for “ T ” replaced by “ S .” Other useful properties of these special functions can be derived from results given by Hobson,²⁹ Erdelyi,³⁰ Abramowitz and Stegun,³¹ and Lebedev.³²

- ¹A. H. Boozer, Phys. Fluids **26**, 1288 (1983).
- ²M. D. Kruskal and R. M. Kulsrud, Phys. Fluids **1**, 265 (1958).
- ³J. M. Greene and J. L. Johnson, Phys. Fluids **5**, 510 (1962).
- ⁴S. Hamada, Nucl. Fusion **2**, 23 (1962).
- ⁵H. R. Lewis, Phys. Fluids B **2**, 2551 (1990).
- ⁶H. R. Lewis, P. G. L. Leach, S. Bouquet, and M. R. Feix, J. Math. Phys. **33**, 591 (1992).
- ⁷B. Abraham-Shrauner and H. R. Lewis, Phys. Fluids B **4**, 2024 (1992).
- ⁸P. M. Morse and H. Feshbach, *Methods of Theoretical Physics* (McGraw-Hill, New York, 1953), p. 1301.
- ⁹J. W. Bates, Ph.D. thesis, Dartmouth College, 1996.
- ¹⁰H. R. Lewis, J. W. Bates, and J. M. Finn, Phys. Lett. A **215**, 160 (1996).
- ¹¹D. W. Kerst, J. Nucl. Energy, Part C **4**, 253 (1962).
- ¹²K. J. Whiteman, Rep. Prog. Phys. **40**, 1033 (1977).
- ¹³L. Turner, J. Math. Phys. **26**, 991 (1985).
- ¹⁴M. S. Janaki and G. Ghosh, J. Phys. A **20**, 3679 (1987).
- ¹⁵M. Tessarotto, J. L. Johnson, and L. J. Zheng, Phys. Plasmas **2**, 4499 (1995).
- ¹⁶Z. Yoshida, Phys. Plasmas **1**, 208 (1994).
- ¹⁷W. D. D’haeseleer, W. N. G. Hitchon, J. D. Callen, and J. L. Shohet, *Flux Coordinates and Magnetic Field Structure* (Springer-Verlag, New York, 1990).
- ¹⁸R. B. White, *Theory of Tokamak Plasmas* (North-Holland, Amsterdam, 1989).
- ¹⁹Y. Tomita, Y. Nomura, H. Momota, and R. Itatani, J. Phys. Soc. Jpn. **44**, 637 (1978).
- ²⁰J. R. Cary and J. D. Hanson, Phys. Fluids **29**, 2464 (1986).
- ²¹G. Gnudi and T. Hatori, J. Phys. Soc. Jpn. **62**, 2030 (1993).
- ²²A. Montavi, K. Graf Finck von Finckenstein, and D. F. Duchs (private communication, 1992).
- ²³J. R. Cary, Phys. Rev. Lett. **49**, 276 (1982).
- ²⁴W. K. H. Panofsky and M. Phillips, *Classical Electricity and Magnetism* (Addison-Wesley, Cambridge, MA, 1955), p. 116.
- ²⁵R. P. Freis, C. W. Hartman, F. M. Hamzeh, and A. J. Lichtenberg, Nucl. Fusion **13**, 533 (1973).
- ²⁶J. W. Bates and H. R. Lewis, “Topological analysis of a perturbed MHD equilibrium using magnetic field line invariants,” to appear in J. Plasma Phys. (1997).
- ²⁷J. R. Cary, Phys. Fluids **27**, 119 (1984).
- ²⁸J. D. Jackson, *Classical Electrodynamics*, 2nd ed. (Wiley, New York, 1975), p. 176.
- ²⁹E. W. Hobson, *The Theory of Spherical and Ellipsoidal Harmonics* (Cambridge University Press, Cambridge, 1931).
- ³⁰A. Erdelyi, in *Higher Transcendental Functions* (McGraw-Hill, New York, 1953), Chap. III.
- ³¹M. Abramowitz and I. A. Stegun, in *Handbook of Mathematical Functions*, 9th ed. (Dover, New York, 1970), Chap. 20.
- ³²N. N. Lebedev, *Special Functions & Their Application* (Dover, New York, 1972).